

Reviewer Documentation

Cover & Executive Summary

ANAHANAD STUDIO

Music School Management System (ERP)

Project Documentation & Technical Portfolio

Prepared by

Sabyasachi Sahani

Junior Full Stack Developer

Technology Stack

React • TypeScript • Refine • Express • Better Auth • Drizzle ORM • PostgreSQL (Neon) •
Cloudinary • Railway • Vercel • Arcjet

Project Links

Source Code

<https://github.com/sibuspd/Anahanad-Studio>

Live Application

<https://anahanad-studio.vercel.app/>

Documentation Version

Version 1.0

Last Updated

July 2026

Reviewer Documentation

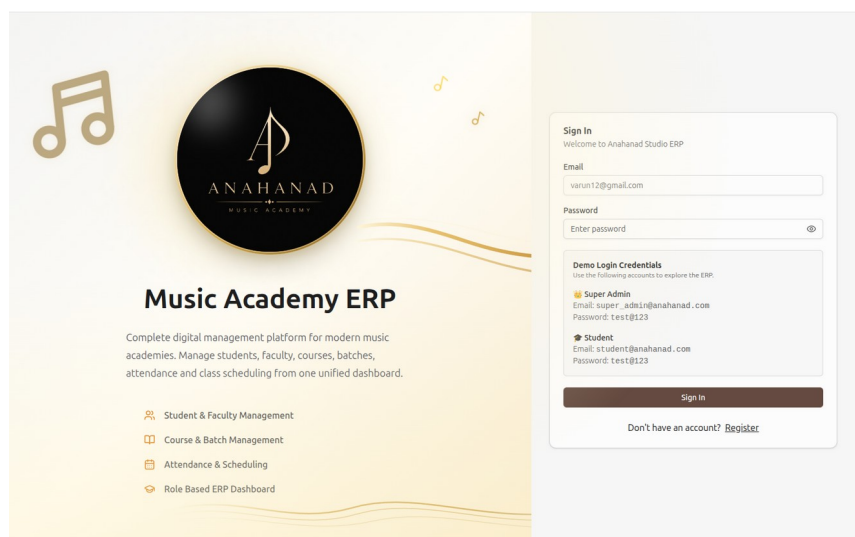
Executive Summary

Anahanad Studio is a modern web-based Enterprise Resource Planning (ERP) application developed to streamline the day-to-day operations of a music academy. The system centralizes administrative workflows by providing secure authentication, role-based access control, academic resource management, and scalable backend services through a modern full-stack architecture.

The application demonstrates practical implementation of contemporary web development technologies including React, TypeScript, Express, PostgreSQL, Drizzle ORM, Better Auth, and cloud-native deployment using Vercel and Railway. The project emphasizes maintainability, scalability, type safety, and production-ready engineering practices while following modular software design principles.

This document presents the system architecture, implementation details, key features, engineering decisions, technical challenges, deployment strategy, and future roadmap of the application. It is intended to provide reviewers, recruiters, and fellow developers with a comprehensive overview of both the business objectives and the engineering practices employed throughout the development lifecycle.

Category	Details
Project Type	Full Stack Web Application
Domain	Education Technology (Music School ERP)
Frontend	React + TypeScript + Refine
Backend	Express.js REST API
Database	PostgreSQL (Neon)
Authentication	Better Auth
Deployment	Vercel + Railway
Media Storage	Cloudinary
Security	Arcjet + Role-Based Access Control



Reviewer Documentation

Core Functional Modules

The current implementation consists of the following functional modules:

- Secure User Authentication
- Role-Based Access Control (RBAC)
- Dashboard Overview
- Department Management
- Subject Management
- Course Management
- Batch Management
- Session Management
- Cloud-Based Media Management
- Administrative User Interface
- Responsive Web Design

System Architecture

The application follows a modern three-tier architecture consisting of:

Presentation Layer

- React
- TypeScript
- Refine Framework

Application Layer

- Express.js REST APIs
- Better Auth
- Business Logic
- Validation
- Authorization

Data Layer

- PostgreSQL (Neon)
- Drizzle ORM
- Cloudinary Media Storage

Reviewer Documentation

Sign In

Welcome to Anahanad Studio ERP

Email

Password

Demo Login Credentials

Use the following accounts to explore the ERP.

👑 **Super Admin**
Email: super_admin@anahanad.com
Password: test@123

🎓 **Student**
Email: student@anahanad.com
Password: test@123

[Sign In](#)

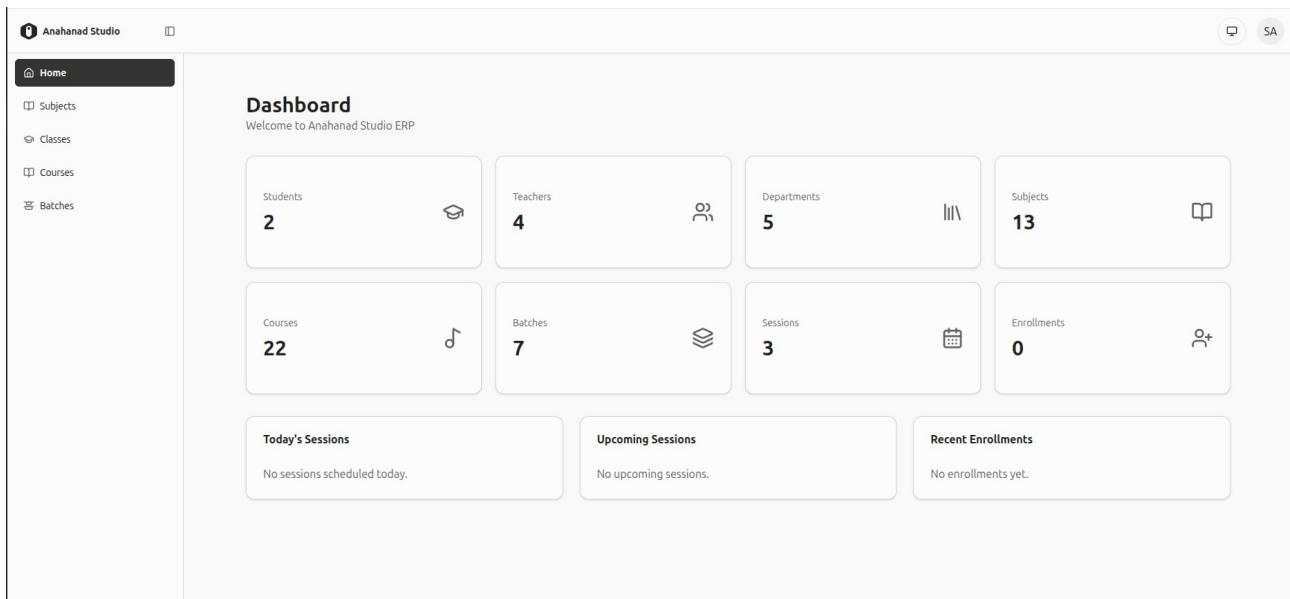
Don't have an account? [Register](#)

To log in, use either of the following demo credentials:

Role	Email	Password	Control
Administrator	super_admin@anahanad.com	test@123	Can Create/Edit/Delete/View
Teacher/User	student@anahanad.com	test@123	Can only View

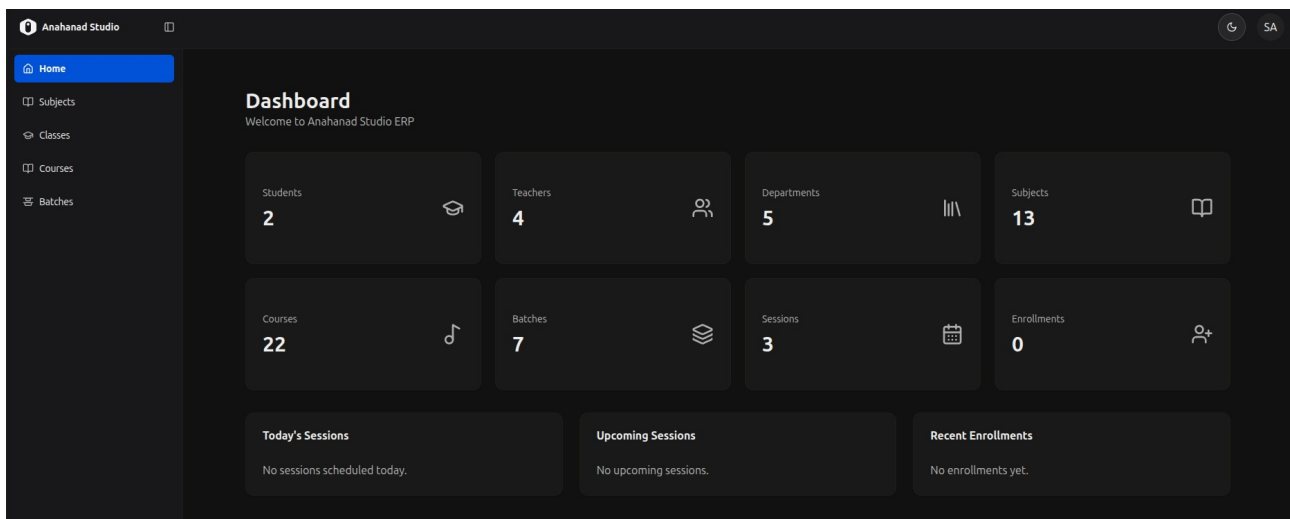
Reviewer Documentation

Dashboard



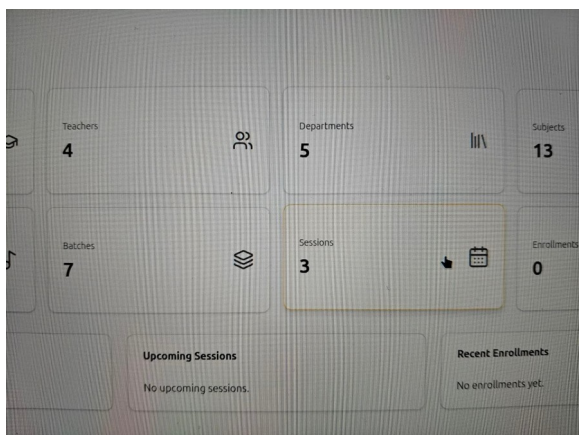
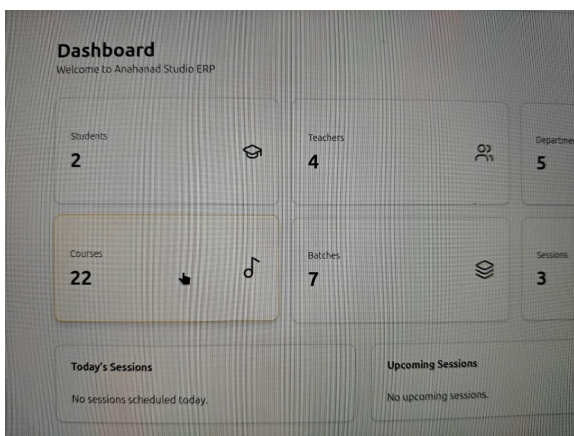
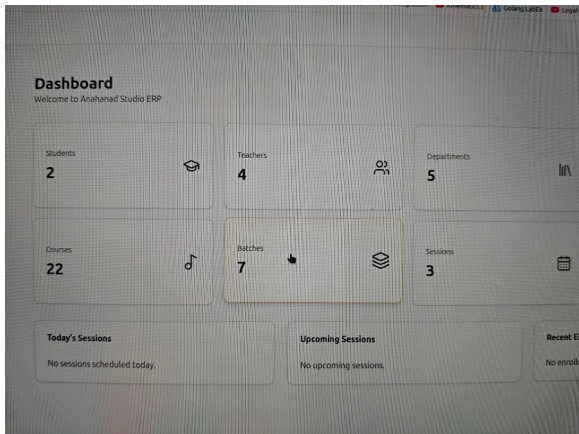
Key Features

- Displays real-time summary statistics for important entities including Students, Teachers, Departments, Subjects, Courses, Batches, Sessions, and Enrollments.
- Provides dedicated dashboard cards that present the total count of records available in the system.
- Displays today's scheduled sessions for quick monitoring of daily academic activities.
- Shows upcoming class sessions to help administrators plan future schedules.
- Displays recent student enrollments for quick administrative reference.
- Includes a responsive sidebar for seamless navigation between different ERP modules.
- Provides quick access to authenticated user profile and application settings.



One-click switching between Light and Dark themes.

Reviewer Documentation



Interactive Dashboard Cards

The dashboard cards act as quick navigation shortcuts to their respective management modules. Each card displays a summary count of records while providing immediate access to detailed management pages through a simple click.

When a user hovers over a card, a visual highlight is applied to indicate that the element is interactive. This improves usability by providing clear feedback before navigation and enhances the overall user experience.

Reviewer Documentation

 > Courses

Courses

Manage all music courses offered by the academy.

Q Search course... All Subjects All Levels

[+ Create](#)

Course	Subject	Level	Duration	Fee	Actions
Intermediate Piano	Piano	intermediate	8 Months	₹ 22000.00	Edit Delete
Advanced Piano Performance	Piano	advanced	12 Months	₹ 35000.00	Edit Delete
Electronic Keyboard Basics	Electronic Keyboard	beginner	6 Months	₹ 14000.00	Edit Delete
Advanced Keyboard Performance	Electronic Keyboard	advanced	12 Months	₹ 32000.00	Edit Delete
Synth Programming	Synthesizer	intermediate	6 Months	₹ 18000.00	Edit Delete
Electronic Music Production	Synthesizer	advanced	12 Months	₹ 36000.00	Edit Delete
Beginner Guitar	Guitar	beginner	6 Months	₹ 15000.00	Edit Delete
Intermediate Guitar	Guitar	intermediate	8 Months	₹ 22000.00	Edit Delete
Advanced Guitar Performance	Guitar	advanced	12 Months	₹ 35000.00	Edit Delete
Beginner Piano	Piano	beginner	6 Months	₹ 15000.00	Edit Delete

Rows per page 10 Page 1 of 3

Course Management – Administrator View (Super Administrator)

 > Courses

Courses

Manage all music courses offered by the academy.

Q Search course... All Subjects All Levels

Course	Subject	Level	Duration	Fee	Actions
Violin Foundation	Violin	beginner	6 Months	₹ 16000.00	
Intermediate Violin	Violin	intermediate	8 Months	₹ 24000.00	
Ukulele Essentials	Ukulele	beginner	4 Months	₹ 12000.00	
Tabla Foundation	Tabla	beginner	6 Months	₹ 15000.00	
Advanced Tabla	Tabla	advanced	12 Months	₹ 30000.00	
Drum Kit Beginner	Drums	beginner	6 Months	₹ 17000.00	
Rock Drumming	Drums	advanced	12 Months	₹ 32000.00	
Cajon Essentials	Cajon	beginner	4 Months	₹ 10000.00	
Western Vocals Beginner	Western Vocals	beginner	6 Months	₹ 15000.00	
Raag Foundation	Hindustani Vocals	beginner	12 Months	₹ 25000.00	

Rows per page 10 Page 2 of 3

Course Management – Restricted User View

The Course Management module enables administrators to manage all music courses offered by the academy. Each course contains structured information such as the associated subject, learning level, course duration, and fee, allowing the institution to maintain a standardized catalogue of training programs. The module supports complete CRUD (Create, Read, Update, Delete) operations for authorized users while ensuring that restricted users have read-only access to course information.

Administrator View

The Super Administrator has full control over course management and can perform all administrative operations through a single interface.

Reviewer Documentation

Restricted User View

For users without administrative privileges, the same interface is presented in a read-only mode. Users can browse available courses and use search and filter functionality, but administrative controls such as **Create**, **Edit**, and **Delete** are intentionally hidden.

Home > Subjects

List Subjects

Quick access to essential metrics and management tools

Search by name

All departments + Create

Code	Name	Department	Description
PIANO	Piano	Keyboard	Western classical and contemporary piano
EKEY	Electronic Keyboard	Keyboard	Electronic keyboard and arranger
SYNTH	Synthesizer	Keyboard	Music production and synthesis
GUITAR	Guitar	Strings	Acoustic and electric guitar
VIOLIN	Violin	Strings	Western and Indian violin
UKULELE	Ukulele	Strings	Ukulele training
TABLA	Tabla	Percussions	Indian Classical Tabla
DRUMS	Drums	Percussions	Western Drum Kit
CAJON	Cajon	Percussions	Cajon percussion
WVOC	Western Vocals	Vocals	Western Vocal Training

Rows per page 10 Page 1 of 2

Home > Subjects

List Subjects

Quick access to essential metrics and management tools

Search by name

Percussions + Create

Code	Name	Department	Description
TABLA	Tabla	Percussions	Indian Classical Tabla
DRUMS	Drums	Percussions	Western Drum Kit
CAJON	Cajon	Percussions	Cajon percussion

Rows per page 10 Page 1 of 1

The Subject Management module provides a centralized view of all musical subjects offered by the academy. Each subject is linked to its respective department and includes essential information such as subject code, subject name, department, and a brief description. This module serves as the foundation for organizing courses and maintaining a structured academic catalogue.

To improve usability, the module includes a department-based filtering feature, allowing administrators to quickly narrow the displayed results and locate subjects belonging to a specific department.

Reviewer Documentation

Key Features

- View all registered music subjects in a structured table.
- Display subject code, name, department, and description.
- Search subjects by name.
- Filter subjects by department.
- Create new subjects (authorized users).
- Paginated subject listing for efficient browsing.
- Responsive table layout for different screen sizes.

Batches

Batches

Manage all teaching batches.

+ Create

Batch	Capacity	Created	Schedule	Actions
Morning Batch	30	03/07/2026	Mon Wed 2 classes/week	<button>Edit</button> <button>Delete</button>
Evening Batch	35	03/07/2026	Tue Thu 2 classes/week	<button>Edit</button> <button>Delete</button>
Weekend Batch	25	03/07/2026	Sat 1 class/week	<button>Edit</button> <button>Delete</button>
Kids Morning Batch	20	03/07/2026	Mon Fri 2 classes/week	<button>Edit</button> <button>Delete</button>
Kids Evening Batch	20	03/07/2026	Tue Thu 2 classes/week	<button>Edit</button> <button>Delete</button>
Working Professionals Batch	18	03/07/2026	Mon Wed 2 classes/week	<button>Edit</button> <button>Delete</button>
Sunday Masterclass Batch	15	03/07/2026	Sun 1 class/week	<button>Edit</button> <button>Delete</button>

Rows per page 10 Page 1 of 1 << < > >>

Batches

Edit

← Edit Batch

Refresh

Batch Information

Batch Name

Weekend Guitar Batch

Capacity

30

Weekly Schedule

Day

Monday

Start Time

09:00

End Time

10:00

Add Schedule

Remove

Update Batch

Reviewer Documentation

Batch Overview

The batch listing provides administrators with a centralized view of all teaching batches available within the academy.

Each record includes:

- Batch Name
- Student Capacity
- Creation Date
- Weekly Teaching Schedule
- Administrative Actions (Edit/Delete)

The schedule is displayed as compact day badges (e.g., **Mon, Wed, Sat**), along with the number of classes conducted each week. This provides an at-a-glance overview of teaching frequency without opening the individual batch details.

Batch Scheduling

Selecting **Edit** opens a dedicated batch configuration page where administrators can modify batch information and manage weekly teaching schedules.

The scheduling interface allows administrators to:

- Update batch information.
- Modify student capacity.
- Assign one or more teaching days.
- Configure class start and end times.
- Add multiple schedule entries for a batch.
- Remove existing schedule entries.
- Save schedule changes through a single update operation.

Session Overview

The session listing provides administrators with a centralized view of all scheduled classes. Each record displays:

- Subject Code
- Course Name
- Department
- Assigned Instructor
- Batch
- Session Date
- Start & End Time
- Current Status
- Unique Invite Code
- Administrative Actions

Reviewer Documentation

The module also includes search and multiple filtering options, allowing administrators to quickly locate sessions based on course, batch, teacher, or session name.

Classes

List of Sessions

Quick access to essential metrics and management tools

Search by Session/Class Name

All Courses

All Batches

All Teachers

+ Create

Subject Code	Course	Department	Teacher	Batch	Date	Time	Status	Invite Code	Actions
GUIAR	Beginner Guitar	Strings	Prasanna Bhure	Kids Morning Batch	12/07/2026	22:41:00 - 02:36:00	scheduled	VBB40TV	View Edit
EKEY	Advanced Keyboar...	Keyboard	Makarand Jadhav	Morning Batch	24/07/2026	22:26:00 - 21:22:00	scheduled	27WUWOA	View Edit
HVOC	Raag Foundation	Vocals	Rishikesh XXX	Evening Batch	08/07/2026	21:08:00 - 23:03:00	scheduled	ERE3Y1D	View Edit

Rows per page 10 Page 1 of 1

Session Details

Selecting the **View** action opens a comprehensive session profile containing detailed academic and administrative information.

Classes > Show

Class Details

Refresh Edit



Introduction to Sitar Strings

STR SCHEDULED

An introductory course in Sitar to get the student used to finger placement on the strings.

INSTRUCTOR

PB

Prasanna Bhure

tereketaka@gmail.com

DEPARTMENT

Strings

Guitar, Violin and Ukulele Department

SUBJECT

Guitar

JOIN CLASS

Reviewer Documentation

Session Name

Introduction to Saptak/Octaves

Description *

Knowing how the pitch level changes and scale rises high or low in a linear progression.

Course Details

Subject

Guitar

Select a batch

Kids Morning Batch

Course

Intermediate Guitar

Assigned Teacher

Select an Instructor

Makarand Jadhav
Prasanna Bhure
Vishwesh XXX
Rishikesh XXX

Status

Scheduled

Session Schedule

Session Date

08/07/2026

Start Time

21:08

End Time

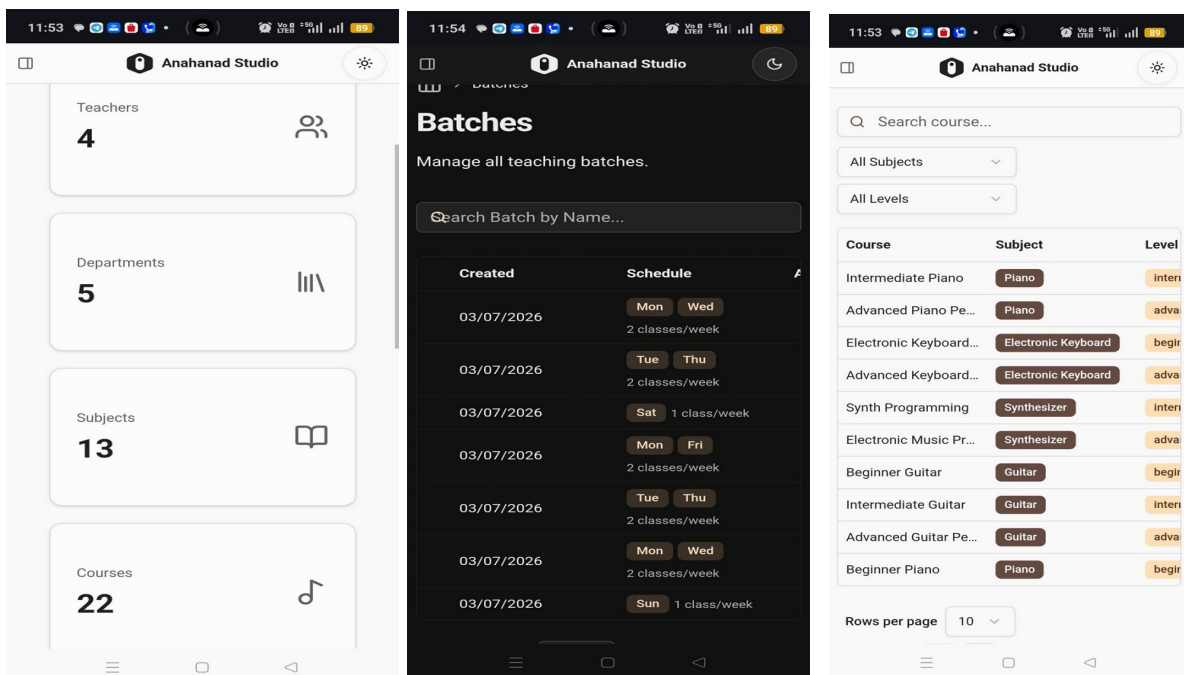
23:03

Cancel

Update Session

Responsive Mobile Experience

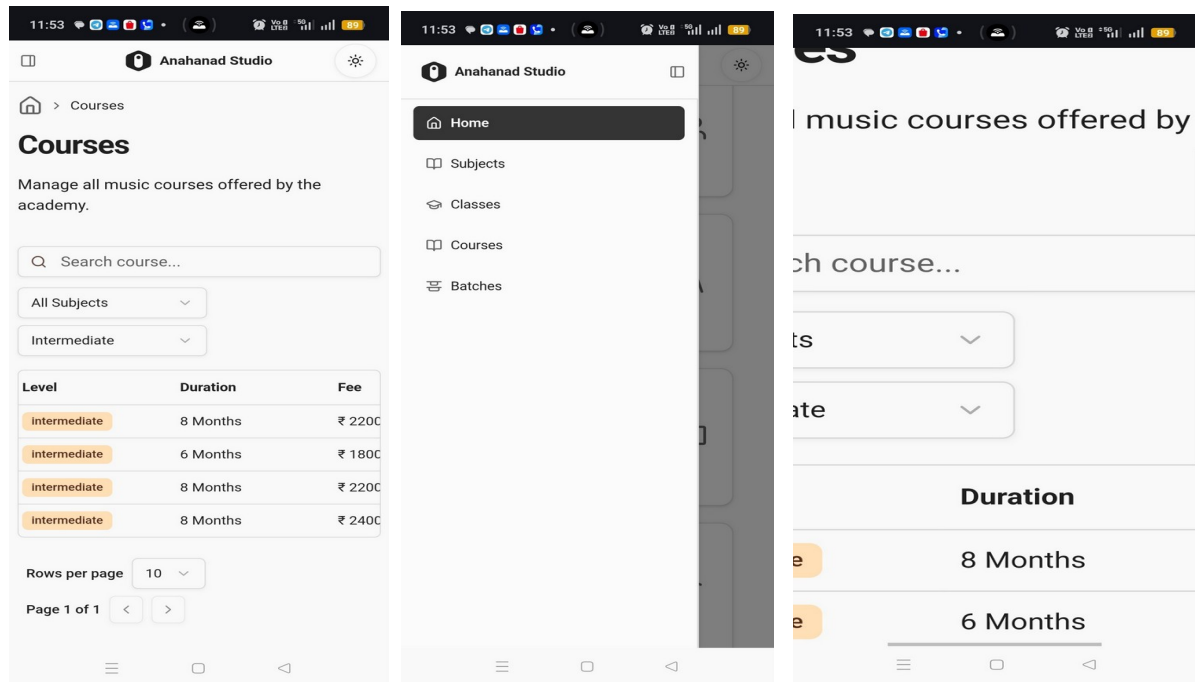
The Anahanad Studio ERP is designed with a mobile-first responsive approach, ensuring that administrators and instructors can access the application from smartphones and tablets without compromising usability. The interface automatically adapts to smaller screen sizes while preserving the functionality available on the desktop version.



Reviewer Documentation

Theme Consistency

Both **Light** and **Dark** themes are fully supported on mobile devices. The selected theme is applied consistently across dashboard widgets, navigation menus, data tables, and management modules, providing a uniform visual experience regardless of screen size.



Final Remarks

Thank you for reviewing the Anahanad Studio ERP project. I hope this documentation provides a clear understanding of the application's architecture, features, and implementation approach. Feedback and suggestions are always welcome as I continue to improve the project and expand its capabilities.

Note: *Anahanad Studio ERP is an actively evolving project. The current version demonstrates the core architecture and primary modules, while several planned features and functional enhancements are still under development and will be introduced in future releases.*